

Pseudonymized but Linkable: Cross-Document Re-identification Risks in LLM-Ready Sensitive Text Corpora

Anonymous authors

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Abstract

1 Sensitive text corpora are commonly transformed one document at a time
2 before retrieval, evaluation, sharing, or model adaptation. This work-
3 flow removes obvious direct identifiers, but it can leave a corpus-level
4 privacy failure mode: records with no names, emails, or account numbers
5 may remain linkable through repeated quasi-identifiers such as role, re-
6 gion, schedule, institution type, family context, health context, financial
7 context, and rare events. We introduce CrossDoc-PrivacyBench, a fully
8 synthetic benchmark with 120 controlled personas, 480 documents, risk
9 tiers, utility labels, and decoy auxiliary profiles. We evaluate document-
10 local transformations under linkage, auxiliary-profile matching, RAG-style
11 exposure, quasi-identifier recovery, and utility metrics. Direct redaction
12 leaves auxiliary-profile matching unchanged at 0.708 top-1, while a Presidio-
13 style PII baseline remains highly matchable at 0.594. We propose Link-
14 Guard, a simple corpus-aware generalization heuristic that estimates cross-
15 document uniqueness and generalizes high-risk quasi-identifiers until a
16 target anonymity set is reached. At target $k = 5$, LinkGuard reduces
17 auxiliary top-1 to 0.042 while preserving issue classification and retrieval
18 Recall@5 at 1.000; aggressive redaction reaches similar privacy but drops
19 issue accuracy to 0.350 and retrieval Recall@5 to 0.312. A cached GPT-
20 5.5 stress audit on 48 T2/T3 personas preserves this ordering, reinforcing
21 that LLM-ready data transformations should be evaluated for corpus-level
22 linkage resistance, not only document-local PII removal.

23 1 Introduction

24 Sensitive data is often most useful when it is not isolated. A hospital may have clinical
25 messages, billing notes, scheduling records, and accommodation requests about the same
26 patient. A legal-aid organization may have intake notes, benefits appeals, landlord disputes,
27 and follow-up emails about the same client. A company may have support tickets, HR
28 notes, finance requests, and internal chat snippets about the same employee or customer.
29 Foundation-model workflows—retrieval-augmented generation (RAG), evaluation-set con-
30 struction, model adaptation, analytics, and data sharing—therefore often operate over
31 *multi-document* corpora rather than one-off documents.

32 The dominant operational response is document-local de-identification. A pipeline removes
33 names, emails, phone numbers, mailing addresses, account numbers, and sometimes other
34 spans that look individually identifying. This is necessary, but it is not sufficient. A
35 rare combination such as a region, occupation, institution type, care schedule, family
36 constraint, legal context, and local event may be harmless in one document yet become
37 a stable signature when repeated across documents. Pseudonymization can make this
38 worse: a stable pseudonym is useful for continuity, but it is also a linkage handle. Per-
39 document pseudonyms remove the explicit handle, but they do not remove the repeated
40 quasi-identifier structure.

41 This paper studies a focused question: *after document-local anonymization or pseudonymization,*
42 *can an attacker still link multiple transformed documents to the same person and match them to aux-*

43 *iliary background profiles?* We intentionally avoid real-person data and real re-identification
 44 attempts. Instead, we build a controlled synthetic benchmark in which every persona,
 45 document, decoy, and utility label is generated by the experimenter. Synthetic construction
 46 is not a substitute for deployment evaluation, but it gives us safe ground truth for controlled
 47 privacy auditing.

48 Our contributions are fourfold. First, we introduce CrossDoc-PrivacyBench, a synthetic
 49 cross-document privacy benchmark with 120 personas, 480 documents, four domains, risk
 50 tiers, utility labels, varied document templates, and decoy auxiliary profiles. Second, we
 51 define a compact audit protocol for multi-document transformations: pairwise linkage,
 52 fixed- K clustering, auxiliary-profile matching, exact quasi-identifier recovery, RAG-style
 53 profile-query exposure, and utility retention. Third, we show that common document-
 54 local transformations leave strong residual risk: direct redaction leaves auxiliary matching
 55 unchanged, a Presidio-style PII baseline remains highly matchable, and stable pseudonyms
 56 make linkage almost trivial. Fourth, we propose LinkGuard, a deliberately simple corpus-
 57 aware generalization heuristic, and show that it can occupy a better privacy-utility point
 58 than blanket aggressive redaction in this synthetic setting.

59 2 Problem Setup and Threat Model

60 **Setting.** A data holder has a sensitive text corpus with multiple records per person,
 61 household, customer, patient, client, or employee. The corpus is transformed before it is
 62 indexed for RAG, shared internally, released for evaluation, or used in model-adaptation
 63 workflows. We consider transformations that operate on text, including direct redaction,
 64 pseudonymization, document-local anonymization, and corpus-aware generalization.

65 **Attacker knowledge.** The attacker sees the transformed corpus. For the auxiliary-profile
 66 task, the attacker also receives a small candidate set containing one true synthetic auxiliary
 67 profile and decoys. This simulates background knowledge such as an external directory,
 68 leaked structured metadata, public biographical context, or another internal table. The at-
 69 tacker does not receive the original direct identifiers, and the candidate labels are neutralized
 70 in the OpenAI audit to avoid leaking synthetic persona IDs.

71 **Attacker goals.** We evaluate three defensive questions. First, can an attacker tell which
 72 transformed documents belong to the same synthetic persona? Second, can the attacker
 73 match a cluster of transformed documents to the correct auxiliary profile among decoys?
 74 Third, can profile-like queries retrieve target documents from a transformed corpus, creating
 75 a RAG exposure channel?

76 **Non-goals and ethics.** We do not collect, process, or attempt to re-identify real peo-
 77 ple. All personas, documents, auxiliary profiles, and decoys are synthetic. The bench-
 78 mark is intended for defensive auditing of transformation methods, not for operational
 79 re-identification. Any extension to real data would require authorization, data minimization,
 80 security review, and domain-specific ethical oversight.

81 3 CrossDoc-PrivacyBench: A Synthetic Cross-Document Benchmark

82 **Personas.** CrossDoc-PrivacyBench contains 120 synthetic personas. Each persona has
 83 direct identifiers—name, email, phone number, address, and account ID—and a structured
 84 set of quasi-identifiers: region, city, occupation, employer type, education, family structure,
 85 medical context, financial context, legal context, schedule pattern, community affiliation,
 86 and rare event. Each persona also has utility labels for four domains. We use deterministic
 87 generation so that experiments are reproducible and no real individual is represented.

88 **Risk tiers.** Personas are assigned to three tiers. T1 documents contain mostly generic
 89 quasi-identifiers, so direct PII is the main privacy risk. T2 documents contain moderately
 90 specific but partly generalized context, such as a city within a region, employer category,

91 broad health or legal context, and schedule constraints. T3 documents contain high-linkage
 92 combinations, such as exact city, occupation, employer type, education, family structure,
 93 treatment or schedule context, affiliation, and rare event. This tiering lets us check whether
 94 an attack and defense are sensitive to increasing quasi-identifier specificity.

95 **Documents.** Each persona has four documents spanning healthcare, legal, financial sup-
 96 port, and HR. The documents are short service or support records with a direct-contact
 97 line and a task-oriented body. Each document includes an issue label, such as medication
 98 refill, benefits appeal, payment deferral, housing issue, accommodation request, or sched-
 99 ule change. The benchmark therefore contains 480 documents total and 384 held-out test
 100 documents from 96 held-out personas.

101 **Auxiliary profiles and decoys.** For each persona, we construct a ten-candidate auxiliary
 102 set. One candidate is the true profile; nine are decoys selected to share some attributes
 103 with the target, such as region, occupation family, or legal context. This makes matching
 104 harder than a trivial exact lookup and lets us report top-1, top-3, and mean reciprocal rank.
 105 The auxiliary-profile task is the central privacy test because it captures both linkage and
 106 background-knowledge use.

107 **Why synthetic data?** Synthetic data makes the evaluation safe, controllable, and repro-
 108 ducible. It also lets us vary quasi-identifier granularity and know which facts should remain
 109 useful for downstream tasks. The tradeoff is external validity: real corpora have richer
 110 style, temporality, omissions, errors, institutional language, and adversary knowledge. We
 111 treat CrossDoc-PrivacyBench as a stress test and design scaffold rather than evidence that a
 112 method is safe for deployment.

113 4 Transformations

114 We compare eight conditions. C0 is the original synthetic corpus. C1 performs direct
 115 redaction of names, first and last names, email addresses, phone numbers, mailing addresses,
 116 and account IDs. C1b applies a Presidio-based PII analyzer plus local regex cleanup for
 117 the synthetic account and address formats. C2 uses stable pseudonyms across the corpus,
 118 replacing direct identifiers and some locations/institutions with consistent tokens. C3 resets
 119 pseudonym mappings per document. C4 is a document-local anonymization proxy that
 120 removes direct identifiers and generalizes selected local spans, such as city and education,
 121 without seeing neighboring documents. C5 is LinkGuard. C6 is aggressive redaction, which
 122 suppresses most potentially identifying context, including quasi-identifiers and task issue
 123 phrases.

124 4.1 LinkGuard: Corpus-Aware Generalization

125 Document-local anonymizers ask whether a span is identifying inside one document. Link-
 126 Guard asks whether repeated details make a persona distinguishable in the corpus. It
 127 operates over persona groups in the synthetic benchmark, but the same logic could be
 128 applied to known customer, patient, case, or account groupings in a controlled deployment.

For each persona and quasi-identifier field, LinkGuard defines a three-level transformation ladder:

exact value $\rightarrow$ semantic category $\rightarrow$ generic suppression.

For example, a city can become a region and then “a broad location”; an occupation can become a job family and then “general worker”; a rare event can become a local-event category and then “a local event.” The method starts from direct redaction and estimates how many personas share the current generalized representation. If the estimated anonymity set is smaller than the target k , LinkGuard selects another field to generalize. The selection score balances rarity, a hand-coded specificity prior, and an approximate utility cost:

$$s(f) = \frac{\text{rarity}(f) + \text{specificity}(f) + 1}{\max(\text{utility_cost}(f), 0.1)}.$$

Fields with rarer values, higher specificity, and lower expected utility cost are generalized first. The process stops when the estimated anonymity set reaches target k or all fields have been generalized or suppressed. This is a heuristic risk-budgeting procedure, not a formal k -anonymity guarantee: the estimate depends on the modeled population, the field schema, and the attacker’s auxiliary knowledge.

5 Attacks and Metrics

Pairwise linkage. We compute signature features over transformed documents, including explicit pseudonym handles and quasi-identifier phrases. Pairwise linkage is evaluated as same-person binary classification over held-out document pairs. We select a threshold on validation personas and report test F1. This metric is especially useful for showing that stable pseudonyms can become linkage handles.

Fixed- K clustering. We also report fixed- K clustering adjusted Rand index, where K is the true number of held-out personas. Threshold graph clustering is brittle outside stable-handle settings, so we treat pairwise linkage and auxiliary matching as the main privacy evidence.

Auxiliary-profile matching. For each held-out persona, the attacker ranks the ten auxiliary candidates using the transformed documents. The main local attacker uses word TF-IDF similarity; robustness checks use character TF-IDF, hybrid TF-IDF, and a field-weighted matcher that explicitly scans exact and generalized schema fields. We report top-1, top-3, and mean reciprocal rank. Lower values indicate better privacy.

Exact quasi-identifier recovery. Auxiliary matching gives an end-to-end privacy signal, but it can hide which information leaked. We therefore scan transformed documents for exact recovery of quasi-identifier fields such as city, occupation, employer type, education, family context, medical context, financial context, legal context, schedule pattern, affiliation, and rare event. This is a structured profile-reconstruction diagnostic.

RAG exposure. To model retrieval exposure, we issue synthetic profile queries against the transformed corpus and measure whether target-persona documents appear in the top retrieved results. We report Hit@5, whether multiple target documents appear in the top ten, document recall at ten, and reciprocal rank. This diagnostic asks whether transformed records remain reachable from profile-like background knowledge even when they contain no direct identifiers.

Utility. We evaluate lightweight task utility with issue classification, task-oriented retrieval Recall@5, and fact preservation. The classifiers and retrievers are local TF-IDF models trained on original synthetic documents. These metrics test whether service-triage information survives transformation; they do not claim to cover all downstream uses.

6 Results

6.1 Main Privacy–Utility Sweep

Table 1 summarizes the main local sweep over 96 held-out personas. Direct redaction leaves auxiliary matching unchanged at 0.708 top-1, confirming that direct PII removal alone does not remove repeated quasi-identifier risk. The Presidio-style baseline improves over direct redaction but remains highly matchable at 0.594. The document-local proxy still leaves auxiliary top-1 at 0.604 and exact quasi-identifier recovery at 0.304. Consistent pseudonymization makes linkage nearly trivial, with pairwise F1 of 0.979, because the stable token itself becomes a cross-document handle.

LinkGuard reduces auxiliary top-1 to 0.042 and exact recovery to 0.000 while preserving issue accuracy and retrieval Recall@5 at 1.000. Aggressive redaction reaches a similar

Cond.	Pair	Aux@1	Exact	Issue	Ret@5	Facts	Edit
C0 Orig	0.963	0.708	0.395	1.000	1.000	0.682	0.000
C1 Redact	0.214	0.708	0.395	1.000	1.000	0.682	0.246
C1b Presidio	0.142	0.594	0.332	1.000	1.000	0.682	0.348
C2 Stable	0.979	0.427	0.273	1.000	1.000	0.679	0.299
C3 PerDoc	0.148	0.427	0.273	1.000	1.000	0.679	0.315
C4 Local	0.155	0.604	0.304	1.000	1.000	0.682	0.288
C5 LinkGuard	0.019	0.042	0.000	1.000	1.000	0.695	0.535
C6 Agg	0.016	0.052	0.000	0.350	0.312	0.389	0.681

Table 1: Main held-out results. Pair is pairwise linkage F1; Aux@1 is auxiliary-profile top-1 accuracy; Exact is exact quasi-identifier recovery; Issue and Ret@5 are utility metrics; Facts is lightweight fact preservation; Edit is character-level edit ratio. Lower Pair, Aux@1, Exact, and Edit are better for privacy or fidelity; higher Issue, Ret@5, and Facts are better for utility.

Cond.	T1 Aux@1	T2 Aux@1	T3 Aux@1	T3 Exact
C1 Redact	0.156	0.969	1.000	1.000
C1b Presidio	0.094	0.688	1.000	0.875
C4 Local	0.156	0.656	1.000	0.818
C5 LinkGuard	0.031	0.000	0.094	0.000
C6 Agg	0.062	0.000	0.094	0.000

Table 2: Risk-tier stratification. Direct and document-local baselines fail most strongly on high-linkage T3 personas; LinkGuard reduces both auxiliary matching and exact quasi-identifier recovery.

175 auxiliary top-1 of 0.052, but issue accuracy drops to 0.350 and retrieval Recall@5 to 0.312.
 176 Thus, in this synthetic sprint, corpus-aware generalization occupies a better privacy-utility
 177 point than blanket suppression. A stricter body-only utility stress test tells the same story:
 178 LinkGuard scores 0.880, direct redaction 0.870, and aggressive redaction 0.438.

179 Bootstrap 95% confidence intervals over held-out personas keep the main contrast sepa-
 180 rated: direct redaction 0.708 [0.615, 0.802], Presidio-style redaction 0.594 [0.490, 0.698], and
 181 LinkGuard 0.042 [0.010, 0.083]. A three-seed no-API sweep preserves the ordering: direct
 182 redaction, Presidio-style redaction, and the document-local proxy have mean Aux@1 of 0.719,
 183 0.628, and 0.663, while LinkGuard averages 0.045. A noisy-style synthetic rerendering lowers
 184 mean template similarity to 0.306 and preserves the ordering: direct/Presidio/document-
 185 local Aux@1 is 0.656/0.562/0.594, while LinkGuard remains at 0.042 with issue/Ret@5 1.000,
 186 and aggressive redaction issue/Ret@5 0.473/0.635.

187 6.2 Risk Tiers and Signal Ablations

188 The benchmark tiers behave as intended. Under direct redaction, Aux@1 rises from 0.156
 189 on T1 to 0.969 on T2 and 1.000 on T3. LinkGuard lowers the corresponding rates to 0.031,
 190 0.000, and 0.094, and reduces T3 exact quasi-identifier recovery to 0.000. The result indicates
 191 that direct redaction is adequate only in the tier where quasi-identifiers were intentionally
 192 generic; it fails as soon as the corpus contains repeated, moderately specific context.

193 Single-signal ablations of direct-redacted documents indicate that role and institution are
 194 the largest local drivers. Removing role and organization drops Aux@1 from 0.708 to 0.500,
 195 while removing location drops it to 0.521. Family, health, rare event, and schedule ablations
 196 have little effect individually in this synthetic corpus, but they still matter in combination
 197 and under the field-weighted stress attacker. This is consistent with the central claim:
 198 linkage risk is often combinatorial, not span-local.

199 6.3 Sensitivity to the Linkage Risk Budget

200 Table 3 varies the target anonymity set size used by LinkGuard. Most of the privacy gain
 201 appears when moving from direct redaction ($k = 1$) to target $k = 2$: Aux@1 drops from 0.708
 202 to 0.062 and exact recovery drops from 0.395 to 0.007. Increasing the target to $k = 5$ removes

k	Min k	Edit	Pair	Aux@1	Field@1	Exact	Issue	Ret@5
1	1	0.246	0.214	0.708	0.979	0.395	1.000	1.000
2	2	0.521	0.036	0.062	0.354	0.007	1.000	1.000
3	3	0.528	0.029	0.052	0.333	0.004	1.000	1.000
5	5	0.535	0.019	0.042	0.240	0.000	1.000	1.000
8	8	0.538	0.017	0.042	0.167	0.000	1.000	1.000
12	12	0.539	0.017	0.042	0.125	0.000	1.000	1.000
20	20	0.544	0.017	0.042	0.104	0.000	1.000	1.000

Table 3: LinkGuard sensitivity to the target anonymity set size. The main auxiliary matcher drops sharply by $k = 2$, while the field-weighted stress attacker continues improving at larger k .

Table 4: GPT-5.5 qualitative evidence extraction over 24 selected synthetic cases. Entries are case-level rates for signal families in the extracted explanations.

Case	n	Role	Loc.	Inst.	High-spec.	Unc.
Direct success	8	0.750	1.000	0.500	0.475	0.000
LG residual	8	0.000	0.000	0.000	0.075	0.875
Agg failure	8	0.000	0.000	0.000	0.179	0.750

exact recovery and lowers the field-weighted stress attacker to 0.240. Larger targets provide diminishing improvements for the main word attacker but continue to reduce field-weighted matching, which falls to 0.104 at $k = 20$. This suggests that corpus-aware generalization can be treated as a tunable risk budget rather than a single fixed transformation.

6.4 Attack Robustness, RAG Exposure, and API Audit

Across word, character, hybrid TF-IDF, and field-weighted auxiliary matchers, direct redaction ranges from 0.510 to 0.979 top-1 and the document-local proxy ranges from 0.479 to 0.979. LinkGuard remains lower, from 0.031 to 0.240, but the field-weighted result exposes residual structured-context risk. The residual analysis finds four LinkGuard top-1 residual matches under the main word attacker; none retain exact quasi-identifier fields, and the median score margin is only 0.0036. This means that LinkGuard does not eliminate all matching signal, but the remaining matches are weak and mostly coarse.

The RAG-style exposure diagnostic shows why linkage evaluation matters for LLM-ready corpora. For high-linkage T3 personas, exact synthetic profile queries retrieve at least one target document in the top five for every direct-redacted, Presidio-redacted, and document-local corpus, and retrieve multiple target documents in the top ten for every such persona. LinkGuard reduces T3 Hit@5 to 0.312 and Multi@10 to 0.000; aggressive redaction reduces Hit@5 further to 0.031 but at much higher utility cost. A deterministic top-5 RAG context-recovery scan on the same T3 profiles shows that direct redaction exposes 9.781 exact quasi-identifier fields on average and the document-local proxy exposes 7.844 exact / 9.469 coarse fields, while LinkGuard exposes 0.000 exact / 0.438 coarse fields. This diagnostic is not a full RAG attack, but it shows that transformed corpora can remain searchable by auxiliary profiles.

Finally, we ran a time-stamped GPT-5.5 auxiliary-matching audit on 48 T2/T3 personas, using synthetic data only, capped calls, cached responses, and store=false. Direct redaction, Presidio redaction, and the document-local proxy remain highly matchable at 1.000, 0.979, and 1.000 top-1. LinkGuard lowers top-1 to 0.438, with top-3 at 0.708 and model-reported uncertainty at 0.917; aggressive redaction is 0.333. A separate 24-person GPT-5.5 document-local anonymization baseline also remains fully matchable at 1.000 top-1, compared with 0.333 for LinkGuard on the same subset. An additional 24-case GPT-5.5 evidence-extraction audit explains the gap: direct-redaction successful matches cite location in 1.000 and role in 0.750 of cases, while LinkGuard residual matches cite role, location, and institution at 0.000 and are uncertain in 0.875. Because API model behavior can change, we treat this as a stress audit rather than the main statistical evidence.

7 Related Work

Classic work on k -anonymity formalized how released records can remain re-identifiable through combinations of quasi-identifiers and motivated generalization and suppression (Sweeney, 2002). Subsequent privacy models such as ℓ -diversity and t -closeness highlighted that anonymity-set size alone is insufficient when sensitive attributes remain homogeneous or distributionally revealing (Machanavajjhala et al., 2007; Li et al., 2007). Sparse-data attacks showed that high-dimensional records can be de-anonymized with limited auxiliary knowledge (Narayanan and Shmatikov, 2008), and later work examined how practical de-identification defenses can fail under background knowledge and downcoding attacks (Cohen, 2022).

Text anonymization benchmarks such as TAB evaluate anonymization decisions and utility in legal text (Pilán et al., 2022). RAT-Bench recently reframes text anonymization around re-identification risk from both direct and indirect identifiers (Krčo et al., 2026). Surveys of text anonymization and anonymization metrics emphasize that privacy, utility, legal requirements, and user-centered expectations are often measured inconsistently (Deußner et al., 2025; Ren et al., 2025). Presidio and OpenAI Privacy Filter illustrate the practical importance of PII detection tools, but such tools are not by themselves complete anonymization guarantees (Microsoft, 2026; OpenAI, 2026).

Recent work on LLM privacy shows that language models can infer personal attributes from text beyond direct memorization (Staab et al., 2024), and that large models can memorize and reveal training data under extraction attacks (Carlini et al., 2021; Nasr et al., 2023). RAG privacy work shows that retrieval databases can leak private records through retrieval and generation channels (Zeng et al., 2024). Autoregressive infilling attacks on de-identified documents further show that background knowledge can help recover masked spans (Charpentier and Lison, 2025). Our contribution is complementary: we focus on cross-document linkage after document-level transformations, where repeated corpus-level quasi-identifiers can survive otherwise reasonable LLM-ready anonymization workflows.

8 Limitations and Ethics

The benchmark is synthetic and template-controlled. This is necessary for safe ground truth and reproducibility, but it limits external validity. Real sensitive corpora may contain richer writing styles, temporal dependencies, institutional terminology, missing fields, contradictions, OCR errors, and adversary knowledge. Our local attacks are lightweight, and the GPT-5.5 audit is a time-stamped API stress test rather than a stable benchmark. LinkGuard is a heuristic generalization method, not a formal privacy guarantee. The field-weighted attacker shows that residual generalized context can still be informative. Utility is measured with issue labels, retrieval, and lightweight fact checks rather than full downstream deployment tasks.

The benchmark is intended for defensive auditing. We do not collect, process, or attempt to re-identify real people. All examples are synthetic and generated from a controlled schema. Any extension to real data would require authorization, data minimization, access controls, logging, and ethical review. Practitioners should not treat successful performance on this benchmark as proof that a real corpus is safe to release or train on.

9 Conclusion

Document-local PII removal is not enough for LLM-ready sensitive corpora. In a multi-document setting, repeated quasi-identifiers can support linkage, auxiliary-profile matching, and profile-query retrieval even after direct identifiers are removed. Our synthetic benchmark suggests that privacy evaluation should include corpus-level linkage resistance, not only span-level PII recall. It also suggests that simple corpus-aware generalization can improve the privacy-utility frontier relative to document-local anonymization and blanket redaction. The broader lesson is that data transformation for foundation models should ask

what a capable attacker can still infer across the corpus, not only what identifying spans were removed from each document.

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